Creating Strings

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1622 | Introductory Problems | https://cses.fi/problemset/task/1622

Given a string, your task is to generate all different strings that can be created using its characters.

Input

The only input line has a string of length n. Each character is between a–z.

Output

First print an integer k: the number of strings. Then print k lines: the strings in alphabetical order.

Constraints

- $1 \le n \le 8$

Example

Input
aabac

Output
20 aaabc aaacb aabac aabca aacab aacba abaac abaca abcaa acaab acaba acbaa baaac baaca bacaa bcaaa caaab caaba cabaa cbaaa